Faithful coordinates and the actual branch radical

November 13, 2026. Symbolic finite transfer [pcc:unverified].

Actual coordinate map. Let e be the actual basis of $K[X]_{<a+1} \times K[X]_{<b+1}$. Its coordinate equivalence gives $E(w)$. The two polynomial projections are precisely the assembled polynomials A_w, B_w in the previous matrix construction. Therefore

$$\deg A_w \le a, \quad \deg B_w \le b, \qquad (A_w = A_z \text{ and } B_w = B_z) \implies w = z.$$

Existing completeness supplies every bounded pair. No random basis, coordinate-independence heuristic or literal matrix rank is assumed.

For actual $S_{ij} = \mu(A_i A_j - f B_i B_j)$ and arbitrary w, z:

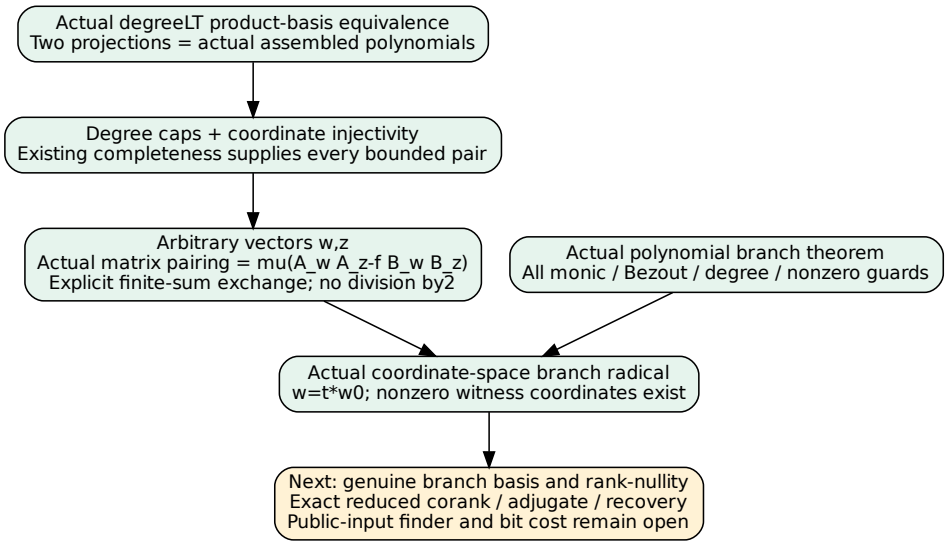
$$\sum_{i,j} S_{ij} w_i z_j = \mu(A_w A_z - f B_w B_z).$$

This expands both vectors and exchanges finite sums. It does not infer a bilinear identity from its diagonal or divide by2.

Actual branch radical. Retain all explicit monic, positive-degree, degree-gap, Bezout and nonzero guards from the polynomial branch theorem. Set $a = \deg F$, $b = \deg H$; the supplied pair is $(\alpha F, \beta H)$. Complete coordinates give a nonzero vector w_0 representing that pair. For every actual branch vector w:

$$\left[\sum_{i,j} S_{ij} w_i z_j = 0 \quad \text{for all branch vectors } z\right] \quad \iff \quad w = t w_0.$$

The proof uses complete test coordinates, the arbitrary-pair identity, the polynomial radical theorem and coordinate injectivity in that order.



Evidence, failed attempts and remaining transfer

Scope. This is the radical of the form restricted to the actual branch hyperplane. It is not the kernel of the full unrestricted matrix, whose polynomial radical has two directions. A reduced branch matrix needs a genuine basis and the remaining rank-nullity/adjugate argument. No rank of an unrelated matrix is asserted. The supplied witness/Bezout data and moments are not found by this theorem; a public-input finder and its bit cost remain open. Jet-family reachability, constant-H and degree-drop charts remain separate.

Formal map. Frozen source `69c822e1c`, `NormMatrixFaithful.lean`; ten preferred declarations cover both actual projections, coordinate injectivity, both assembled degree caps, the arbitrary-vector polynomial double sum, actual matrix bilinear identity, scalar assembly, coordinate branch radical and nonzero witness-coordinate existence. Two original degree-name wrappers preserve the initial statement bindings. Exact hypotheses and current unsigned PCC bindings accompany the Markdown report.

Checks. Pinned Lean4.35.0, CryptoLean148f5dc, Mathlibcfc55fc. Builds007–009 complete3151 jobs. The current tracked-source gate and fresh import-inclusive replay have attributable completion receipts; no additional trusted axiom is admitted. All producer standing remains unsigned.

Failures retained. Draft002 fails on exchanged summation indices; the diagonal would conceal the mismatch. An explicit sum exchange fixes003. Draft004 fails on scalar smul versus field multiplication;005 fixes it. Draft006 adds nonzero witness coordinates. Failed logs are preserved.

The initial ten unsigned PCC bindings were appended before their preservation audit exposed a lookup collision: the maintained tool matches theorem short names, so new lowerDegree shadowed an older theorem despite distinct Lean namespaces. Older source was unchanged. Unique preferred degree names alone were insufficient: retained records also need checking. The canonical lookup now resolves qualified namespaces exactly and rejects ambiguous/unsupported bindings. Original-name wrappers keep initial statement hashes. All127 stored study bindings match;25 PCC tests and62 canary classes pass. Original files, replacement records and append-only notes remain. Shared lock digests are updated; downstream copies follow the canonical owner change. Fresh001 was stopped with143;002 completed before wrappers;003 checks the final source. No signed verdict is invented.

The sparse-workspace source-only check still exits1 on151 other records, versus152 with the old verifier; no new failures. All21 new claims are not_covered without Lean on the default PATH. Separate direct kernel receipts cover the actual modules. No clean global or signed verdict is asserted.

Sources. Pinned Mathlib Basis.equivFun/equivFun_symm_apply and actual degreeLT product basis; existing assembly/moment-matrix definitions, actual polynomial branch theorem and continuation plans. Standard basis/form transfer, no priority claim. KB MCP/wrapper unavailable; bounded fallback. Primary summation-polynomial context remains Galbraith15.8.1–2, source-pinned in the adjacent cost audit.

Next. Choose a genuine basis of the actual branch kernel, identify the reduced matrix kernel with the nonzero witness line, derive exact corank and a nonzero adjugate, then supply every guarded recovery hypothesis.

Resources. Local proof correctness only; no mathematical Python research, new paid job or timing/scaling campaign. Python was used only for maintained PCC administration. Monthly isolab budget remains exhausted. No benchmark, leaderboard or ecdb edit. Canonical measured figures remain unchanged. Editable DOT/SVG and exact receipts accompany this report.